Machine Learning vs. Deep Learning

AIML-500 Machine Learning Fundamentals · 2.3 Assignment

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Introduction

The choice between machine learning (ML) and deep learning (DL) mainly depends on the type of data and the problem. Traditional ML works well when the important features are already known and the data is structured (like tables). DL works better when the data is raw and unstructured, such as images or text, and the useful patterns are too complex for people to define by hand.

This report looks at one example of each approach from the lesson:

1. Machine learning: Customer churn prediction with a Support Vector Machine (SVM), used by a telecom company to keep customers.
2. Deep learning: Image recognition with a Convolutional Neural Network (CNN), used in Apple's Face ID.

For each example, I explain why the chosen approach is suitable and why the other approach is not.

Example 1 (Machine Learning): Customer Churn Prediction

In the lesson's example, a telecom company uses an SVM to predict whether a customer is likely to cancel their service. The model learns from historical customer records using features such as customer age, contract length, and monthly charges. Each record is labeled as "churned" or "stayed."

This is a real problem in the telecom industry. A 2022 study in *PLOS ONE* reported that the industry's average annual churn rate is 20–40%, which can cause large revenue losses (Nguyen et al., 2022). The researchers built a churn model using a kernel SVM on a dataset of 3,333 telecom customers with 19 features. After selecting useful features and balancing the classes, the model achieved very high accuracy (around 99%) and a high F1-score. A telecom company can use this kind of prediction to identify at-risk customers early and offer them better plans before they leave.

Why machine learning is suitable

- The data is structured. Churn data is in a table format, with one row per customer and clear features like contract length and monthly charges. Analysts can choose these features based on business knowledge, which matches the traditional ML approach of manual feature selection.
- The dataset size is moderate. The study used only 3,333 records, but the SVM still performed well. Traditional ML can work effectively with smaller datasets and less computing power.

- The results need to be explainable. Managers need to understand why a customer is at risk so they can take action. Because people choose the features, the company can see which factors (such as high bills) are related to churn.
- It is fast and relatively inexpensive. An SVM can be trained quickly on a regular computer and is easy to update when new data comes in.

Why deep learning is not suitable

- It usually needs much more data. Deep learning generally requires large datasets. With only a few thousand records, and only about 14% of them being churners (Nguyen et al., 2022), a deep neural network could easily overfit.
- There are no complex hidden features to discover. Deep learning's main strength is learning features automatically from raw data (like text). In churn prediction, the features are already clear and meaningful, so this advantage is not very useful.
- It is harder to interpret. A deep network is often like a "black box," which makes it difficult to explain to managers why a specific customer was flagged as high risk.
- It costs more for limited benefit. Deep learning needs more computing power and longer training. Since the SVM already reached very high accuracy, the extra cost would bring little or no real benefit.

Example 2 (Deep Learning): Image Recognition for Facial Recognition

In the lesson's example, a CNN learns to recognize faces directly from raw pixel data. It does not need predefined features such as the shape of the eyes or nose. The network learns patterns step by step, from edges to textures to complete facial features.

A well-known real-world application is Apple's Face ID, which is used to unlock iPhones and approve Apple Pay payments. Face ID uses neural networks trained for facial matching. Apple trained these networks on over a billion images, including infrared and depth images. A separate network is also trained to detect spoofing attempts (such as photos or masks). Face ID can also update the stored face data over time, so it still works after changes like growing a beard or wearing makeup (Apple, n.d.). Facial recognition like this is also widely used for biometric login and security checks (AIMultiple, 2026).

Why deep learning is suitable

- The input is unstructured data. A face image is made of thousands of pixel values. Individual pixels do not have meaning by themselves; the meaning comes from complex patterns. CNNs can learn these patterns

automatically through multiple layers.

- The patterns are too complex to define by hand. A person's face can look different because of lighting, angle, glasses, facial hair, or aging. It is not practical to write manual rules for all these situations. Deep learning can learn these variations from examples.
- A very large amount of training data is available. Deep learning needs big datasets to work well. Apple trained its face-matching networks on over a billion images (Apple, n.d.), so this requirement is clearly met.
- High accuracy is important. Face ID protects personal data. Deep learning can learn subtle details and detect fake attempts, which is necessary for security.

Why traditional machine learning is not suitable

- It depends on manual feature engineering. An SVM or decision tree would need people to define features first, such as the distance between the eyes or the width of the nose. These hand-made features are fragile and do not work well when the head angle, lighting, or expression changes.
- It struggles with high-dimensional data. A single image has thousands of raw values. Traditional ML is not good at learning the spatial relationships between pixels the way CNN layers do.
- It does not improve as much with more data. Traditional ML performance often stops improving after a certain point. Deep learning can keep improving as more training images are added, which is useful when there is a very large dataset.
- It is harder to meet security requirements. A weaker model would have more false unlocks or rejections and would be easier to fool with a photo, which is not acceptable for a system that protects payments.

Comparison

Factor	Churn prediction (ML)	Facial recognition (DL)
Data type	Structured table	Unstructured images (raw pixels)
Feature extraction	Chosen by people	Learned automatically
Data size	Small to moderate	Very large
Computing power	Ordinary computer	Specialized hardware
Explainability	High	Low (acceptable when accuracy is most important)

Main priority	Fast, low-cost, explainable predictions	High accuracy and security

Conclusion

Neither approach is always better; the best choice depends on the problem. Customer churn prediction is suitable for traditional ML because the data is structured, the important features are known, and the business needs explainable results from a moderate amount of data. Facial recognition is suitable for deep learning because the data is raw and unstructured, the patterns are too complex to define by hand, and large datasets and strong hardware are available.

The lesson's summary describes this difference well: machine learning is like giving a student a checklist of things to look for, while deep learning is like teaching the student to read and letting them figure out the rules from experience. Churn prediction needs the checklist. Facial recognition needs the student who learns from experience.

References

AIML-500. (n.d.). *Machine learning vs. deep learning* [Course lesson].

AIMultiple. (2026). *Top deep learning use cases & with examples*.From https://aimultiple.com/deep-learning-applications

Apple Inc. (n.d.). *Facial matching security* [Apple Platform Security Guide]. From https://support.apple.com/guide/security/sece151358d1/web

Nguyen, N. Y., Tran, V. L., & Dao, V. T. S. (2022). Churn prediction in telecommunication industry using kernel Support Vector Machines. *PLOS ONE, 17*(5), e0267935. https://doi.org/10.1371/journal.pone.0267935